# Using Calculus in 2D

Exam code: 9MA0

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Updated on 25 November 2025

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## Using calculus in 2D

### Which quantities in kinematics are vectors?

- It is important you understand **Calculus in 1D** first
- Also recall the differences between key **vector** and **scalar** quantities
- **Displacement** - from starting point
- **Distance** – from start/origin/another particle (all could be different)
- **Speed** and **Velocity**
- **acceleration** and **magnitude** of **acceleration/deceleration**
- **time** is a scalar quantity

### How do I use calculus for 2D vectors?

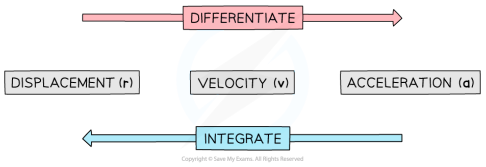
- To **differentiate** a vector:
  - Differentiate both (**i** and **j**) components of the vector
- **To integrate** a vector:
  - Integrate both (**i** and **j**) components of the vector
  - There will be a constant of integration, **c**, which will be a vector made up of two values (an **i**-component constant and a **j**-component constant)
  - Find **c** by substituting in any known vectors, in the same way as you would if it were in 1D

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

## How is differentiation and integration related to displacement, velocity and acceleration?

- In 2D we normally use the vector **r** to represent displacement instead of s
- Just like in 1D:
  - **Differentiate** displacement to get velocity and velocity to get acceleration
  - **Integrate** acceleration to get velocity and velocity to get displacement




## How can 2D-vector questions be made harder?

- Harder problems could involve:
  - using Newton's Second Law (N2L) **F** = *m***a**
  - two vectors being equal requiring both **i** and **j** components to be equal
  - exponentials, logarithms, trigonometric functions (not just polynomials)

- suvat equations (in vector form) would only be involved if **acceleration is constant**
- You may have to find the **magnitude** of the vectors
  - The magnitude of the **velocity** is the **speed**
  - The magnitude of the **displacement** is the **distance** from the **starting point**

- The position vector of a particle is $\mathbf{r} = \mathbf{r}_0 + \mathbf{s}$
  - $\mathbf{r}_0$ is the **initial position** of the particle
  - **s** is the **displacement** of the particle from its initial position

### Worked Example

A ship moves with velocity, **v** m s$^{-1}$, where $\mathbf{v} = (2t)\mathbf{i} + (e^{0.5t})\mathbf{j}$. Initially the ship has position vector $(3\mathbf{i} - 4\mathbf{j})$ m from a fixed origin O.

(a) Find an expression for the position vector of the ship after *t* seconds.

**Answer:**

$$r = \int v \, dt$$

$$r = \int (2t)\mathbf{i} + (e^{0.5t})\mathbf{j} \, dt$$

$$r = (t^2)\mathbf{i} + (2e^{0.5t})\mathbf{j} + \mathbf{c}$$

Remember constant of integration is a vector

Step 3: Find "+c"

$t = 0 \quad r = 3\mathbf{i} - 4\mathbf{j}$

$$3\mathbf{i} - 4\mathbf{j} = 0\mathbf{i} + 2\mathbf{j} + \mathbf{c}$$

$$\mathbf{c} = 3\mathbf{i} - 4\mathbf{j} - 2\mathbf{j}$$

$$\mathbf{c} = 3\mathbf{i} - 6\mathbf{j}$$

$$\mathbf{r} = ((t^2 + 3)\mathbf{i} + (2e^{0.5t} - 6)\mathbf{j})\text{m}$$

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(b) Find the magnitude of the acceleration when $t = 3$ .

**Answer:**

b) Step 1: Identify whether to differentiate or integrate

Acceleration is the derivative of velocity

Step 2: Differentiate velocity

$$\mathbf{a} = \frac{d\mathbf{v}}{dt}$$

$$\mathbf{a} = \frac{d}{dt}((2t)\mathbf{i} + (e^{0.5t})\mathbf{j})$$

$$\mathbf{a} = 2\mathbf{i} + (0.5e^{0.5t})\mathbf{j}$$

Step 3: Find acceleration at the stated time

$t = 3 \quad \mathbf{a} = 2\mathbf{i} + 0.5e^{1.5}\mathbf{j}$

Step 4: Find the magnitude

$$|\mathbf{a}| = \sqrt{2^2 + (0.5e^{1.5})^2}$$

$$|\mathbf{a}| = 3.0035...$$

$$|\mathbf{a}| = 3.00 \text{ ms}^{-2} \text{ (3 sf)}$$

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## Examiner Tips and Tricks

- If the question refers to the direction that the particle is travelling, then you would use the velocity. If the **direction of motion** is asked for then it should be clear from the question whether they want an angle, bearing or vector which it is parallel to.

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